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Department of Electronics and Communication Engineering
Academic Year: 2022- 2023 (Even Semester)

INNOVATIVE TEACHING METHOD

Degree, Semester& Branch: IV Semester B.E. ECE A
Course Code & Title: EC3492 Digital Signal Processing
Name of the Faculty member: Mrs.G.Gnana Priya

Sl.No.	Topic(s)	Activity	Reference(s)
UNIT I - DISCRETE FOURIER TRANSFORM			
1.	Linear filtering using FFT, Revision-Unit I	One Minute Paper	Discrete-Time Signal Processing, A. V. Oppenheim, R.W. Schafer and J.R. Buck, Pearson, 2004.

Sl.No.	Topic(s)	Activity	Reference(s)
UNIT II INFINITE IMPULSE RESPONSE FILTERS			
2.	Cascade, parallel realizations	Brain Storming	https://nptel.ac.in/courses/117102060 Digital Signal Processing, IIT Delhi, Prof.S.C.Dutta Roy

UNIT III FINITE IMPULSE RESPONSE FILTERS			
3.	FIR filter structures - direct form realizationsl	Zero Minute Speech	https://nptel.ac.in/courses/108106151 Digital Signal Processing, IIT Madras, Prof. C. S. Ramalingam

Signature of the Faculty member

HOD

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Innovative Practice Description

- **Unit / Topic:** Unit V / DSP Architecture– Fixed point architecture, Floating point architecture principles
- **Course Outcome:** CO5
- **Unit Outcome:** 5c
- **Activity Chosen:** Flipped Classroom
- **Justification:**

  The Unit V consists of applications of DSP. The concept of DSP processor is very important. So I taught the introduction to DSP processor in the class and then planned to conduct the principles of Fixed point architecture and Floating point architecture as a Flipped Classroom activity to make the students understand the two concepts clearly which enhances their learning level. From their presentation I can observe the understanding level of the students.
- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

  The materials related to the activity were given to the students for learning. The activity was implemented by forming teams. The team forming choice was given to the students with the condition that a team should consist of mixer of advanced learners, average learners and slow learners. In such a way the class strength was divided into 10 teams. Each team prepared and presented Fixed point architecture and Floating point architecture as a Flipped Classroom activity. The first 5 teams presented Fixed point architecture concept and the next 5 teams presented Floating point architecture concept.

CO – PO / PSO mapping:

CO	PO1	PO2	PO8	PO9	PO10	PSO2	PSO3
CO2	2	2	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO8	PO9	PO10	PSO2	PSO3
	2	2	2	2	2	1	1
Justification for correlation	DSP Processors requires the knowledge of engineering fundamentals such as processor flow, Accumulators, storage devices. This course outcome is highly correlated.	Multirate signal processing needs to identify the engineering and other relevant knowledge that applies to a given problem. So, the course outcome is highly correlated.	Flipped Class activity involves presentation which apply moral & ethical principles to known case studies while collecting materials related to the topic. The course outcome is moderately correlated.	Flipped class activity needs to demonstrate effective communication, problem-solving, conflict resolution and leadership skills and listen to other members. The course outcome is highly correlated.	The topic Fixed point architecture principles deals with presentations and preparation of contents which requires written skills. This task is accomplished through Flipped class activity therefore course outcome is correlated highly.	The course outcome is moderately corrected because the concept of FFT using DSP processor is required in design of ARM processors.	The knowle of D processors its architec is useful designing VLS and Communication systems. The course outcome is hi correlated.

- **Images / Screenshot of the practice:**

 

- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

The following points were observed based on the feedback got from the students.

- ✓ The students felt that the self-learning capability improved because of this activity.
- ✓ The presentation skills also increased while learning the concept by themselves and explaining to the class.
- ✓ The communication skills of the students improved while discussing with their team members and then presenting to the class.

❖ ***Benefit of the practice:***

- ✓ The main benefit of the flipped class activity is each group will learn one architecture and the presentation helps the other members also to know about the other architecture.
- ✓ The students effectively involved in the activity by discussing with their team members.
- ✓ The understanding level and learning level of the students also increased.
- ✓ As the team forming choice was given to the students they actively participated.

❖ ***Challenges faced in implementation:***

- The biggest challenge was making the students to learn the materials. Each student has to read the material posted in the canvas and then they have to prepare.
- Very few students were not actively participated in the activity. Making all the students to participate was also challenging.
- The activity was planned for one hour but it took extra time than the allotted time to complete the activity.

References:

❖ A. V. Oppenheim, R.W. Schafer and J.R. Buck, —Discrete-Time Signal Processingl, 8th Indian Reprint, Pearson, 2004.

❖ John G. Proakis & Dimitris G.Manolakis, —Digital Signal Processing – Principles, Algorithms & Applications, Fourth Edition, Pearson Education / Prentice Hall, 2007.

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Signature of the Faculty member **HOD**